5 $G(S) = \frac{10+12}{S(S+2)}$



$H(s) = 1$

a) 1) System type: 1

2) $k_p = G(0) = \frac{22}{0(0+2)} = \infty$

$e_{ss} = \frac{1}{1+k_p} = \boxed{0}$

3) $k_v = \lim_{S \to 0} S\,G(S) = \frac{22S}{S(S+2)}\Big|_{S=0} = \frac{22}{0+2} = 11$

$e_{ss} = \frac{1}{k_v} = \frac{1}{11} = \boxed{0.09}$ percentage is 9%

4) $k_a = \lim_{S \to 0} S^2 G(S) = \frac{22S^2}{S(S+2)}\Big|_{S=0} = \boxed{0}$

$e_{ss} = 1/k_a = 1/0 = \boxed{\infty}$

b)